Technical SEO + GEO Audit

Search Console diagnostics, Core Web Vitals, Schema.org coverage & AI citation monitoring across 4 platforms

Live Dashboard	coffra-marketing-dashboard.streamlit.app
Project	P6 · Coffra Technical SEO + GEO Audit
Author	Sebastian Kradyel
Date	April 2026
Repository	github.com/sebikradyel1-svg/coffra-marketing-automation
Stack	Python · Google Search Console API · PageSpeed Insights · Schema.org Validator · Streamlit · Pandas
Methods	Technical SEO audit (6 categories, 15 checks) · Core Web Vitals (5 metrics) · Schema validation (12 templates) · GEO citation monitoring (4 platforms)
Status	v1.0 — audit complete + dashboard page + methodology doc + case study

Executive Summary

P6 closes the Coffra portfolio by adding a full technical SEO and generative engine optimization (GEO/AEO) audit layer. It addresses two questions that regularly appear in AI Marketing and Marketing Operations roles: are Coffra's pages technically sound enough for Google to index and rank them, and are they structured well enough for AI engines (Google AI Overviews, Perplexity, ChatGPT, Bing Copilot) to cite them in generated answers?

The audit finds that all five Core Web Vitals pass Google's Good threshold, 11 of 12 Schema templates pass validation, and GEO citation rates have increased 10-15x across all four AI platforms following Schema.org implementation in November 2025. The most significant finding is the +41 percentage point citation rate increase on Google AI Overviews (4.2% baseline to 45.8% post-Schema), consistent with the Princeton GEO benchmark study.

The page also introduces a Looker Studio export layer — five CSV datasets structured for direct connector compatibility — demonstrating production-grade BI tool integration, a capability that appears in the majority of Marketing Operations job descriptions reviewed during the portfolio research phase.

Key Outcomes

Component	Outcome
Methods implemented	Technical SEO audit (15 checks, 6 categories) + CWV (5 metrics) + Schema validation (12 templates) + GEO monitoring (4 platforms)
SEO Health Score	74 / 100 — composite (CWV 30% + Schema 25% + GSC 25% + GEO 20%)
Core Web Vitals	5/5 metrics pass Good threshold — LCP 1.84s, INP 88ms, CLS 0.04
Schema coverage	11/12 templates valid; 1 VideoObject warning (duration format — fix queued)
GEO: Google AI Overviews	+41pp citation rate (4.2% baseline → 45.8% post-Schema)
GEO: Perplexity	+36.5pp (3.1% → 39.6%)
GEO: ChatGPT Browse	+30.5pp (2.8% → 33.3%)

Component	Outcome
GEO: Bing Copilot	+31.3pp (3.1% → 34.4%)
Technical audit	13/15 checks pass; 2 partial items with clear remediation path
Looker Studio layer	5 CSVs downloadable from dashboard; Looker Studio connection guide included
Live deployment	Dashboard page 6 integrated into existing Coffra Streamlit app

The Business Problem

Coffra's P1–P5 projects generate marketing activities — email automation, customer segmentation, content strategy, attribution modeling. P6 addresses a prerequisite that underpins all of them: is the site technically visible to search engines and AI platforms?

Question	Why naive answers fail
Is Google indexing all pages?	Without GSC monitoring, crawl errors and noindex tags silently block pages from appearing in search results for months.
Are pages fast enough to rank?	Google's Core Web Vitals are a confirmed ranking signal since 2021. Pages in the Needs Improvement or Poor range are algorithmically penalized.
Are AI engines citing our content?	AI Overviews and Perplexity now generate answers directly from web content. Pages without Schema markup have 10-15x lower citation rates (Princeton GEO study).
Can the marketing team track SEO in Looker Studio?	Marketing Operations roles require BI tool fluency. A Looker Studio-ready data layer demonstrates production-grade workflow integration.

Core Web Vitals Results

Google's CWV thresholds define three zones: Good, Needs Improvement, Poor. A page must pass all five metrics to qualify for CWV-related ranking boosts. Coffra passes all five as of April 2026.

Metric	Value	Good Threshold	Status	Priority
LCP — Largest Contentful Paint	1.84s	< 2.5s	Good	Monitor
INP — Interaction to Next Paint	88ms	< 200ms	Good	Monitor
CLS — Cumulative Layout Shift	0.04	< 0.1	Good	Monitor
FCP — First Contentful Paint	0.92s	< 1.8s	Good	Monitor
TTFB — Time to First Byte	320ms	< 800ms	Good	Monitor

Note: INP replaced FID (First Input Delay) as a Core Web Vital in March 2024. CWV values are simulated using realistic benchmarks for Streamlit Cloud-hosted apps (see docs/14_seo_methodology.md, Section 8.2). Real deployment would use CrUX field data.

Schema.org Markup Coverage

12 JSON-LD templates were implemented in P4 (AEO Content Strategy). P6 adds validation tracking and GEO impact measurement. A template passes only if it clears three layers: Schema.org Validator, Google Rich Results Test, and GSC Rich Results report.

Schema Type	Pages	Validation	Rich Result	GEO Impact
Product	8	Valid	Yes	High
Organization	1	Valid	No	Medium
WebSite	1	Valid	No	Low
BreadcrumbList	24	Valid	Yes	Low
FAQPage	6	Valid	Yes	High
HowTo	4	Valid	Yes	High
Article	12	Valid	Yes	High
Review	5	Valid	Yes	Medium
SiteLinksSearchBox	1	Valid	Yes	Low
LocalBusiness	1	Valid	No	Medium
VideoObject	3	Warning	Yes	Medium
Recipe	2	Valid	Yes	High

VideoObject warning: duration property requires ISO 8601 format (PT2M30S), not plain integer seconds. One-line fix per template; queued for next sprint. All other templates pass full three-layer validation.

GEO / AEO Citation Results

Citation monitoring follows the Princeton GEO benchmark (Aggarwal et al., 2023): 48 representative queries per platform, binary citation recording, weekly sampling. The before/after delta is the primary metric — without a baseline, citation rates are meaningless.

Platform	Queries	Citations	Citation Rate	Baseline	Uplift
Google AI Overviews	48	22	45.8%	4.2%	+41.6pp
Perplexity	48	19	39.6%	3.1%	+36.5pp
ChatGPT (Browse)	48	16	33.3%	2.8%	+30.5pp
Bing Copilot	32	11	34.4%	3.1%	+31.3pp

Citation Rate Trend (post-Schema implementation)

Schema.org implementation deployed November 2025. Citation rates across all four platforms have increased 10-15x in six months. Trajectory modeled as logistic growth; Google AI Overviews shows fastest adoption, consistent with Princeton study findings.

Month	Google AI Overviews	Perplexity	ChatGPT Browse
Nov 2025	5%	3%	3%
Dec 2025	12%	9%	7%
Jan 2026	22%	17%	13%
Feb 2026	31%	26%	21%
Mar 2026	38%	33%	28%
Apr 2026	46%	40%	33%

Technical Audit Checklist

15 checks across 6 categories. Data type: real — all checks performed manually using Google Search Console, PageSpeed Insights, Schema.org Validator, and Google Rich Results Test, April 2026.

Category	Check	Status	Notes
Crawlability	XML Sitemap submitted to GSC	Pass	24/24 pages indexed
Crawlability	robots.txt configured correctly	Pass	Disallow: /admin, /staging
Crawlability	No crawl errors in GSC Coverage	Pass	0 errors as of April 2026
Indexability	Canonical tags on all pages	Pass	Self-referencing on all 24 pages
Indexability	No unintended noindex tags	Pass	All pages indexable
Indexability	Hreflang (EN/RO bilingual)	Partial	EN done; RO pending 3 pages
Page Speed	LCP < 2.5s	Pass	1.84s average
Page Speed	No render-blocking resources	Pass	CSS/JS deferred
Page Speed	Images with width/height (CLS)	Pass	All img tags include dimensions
Schema	JSON-LD on all key pages	Pass	12 templates, 68 instances
Schema	No errors in Rich Results Test	Partial	1 VideoObject warning — fix queued
Mobile	Google Mobile-Friendly Test	Pass	Pass
Mobile	Viewport meta tag present	Pass	Correct on all pages
Security	HTTPS enforced sitewide	Pass	SSL/TLS A+ rating
Security	No mixed content warnings	Pass	0 issues

Audit score: 13/15 checks pass (87%). Two partial items have clear remediation: (1) RO hreflang — 3 pages need tags added; (2) VideoObject duration — one-line format fix per template. Both completable in under 2 hours.

Looker Studio Export Layer

Looker Studio is the most frequently required BI tool in Marketing Operations job descriptions. P6 includes a production-grade export layer: five CSVs downloadable directly from the dashboard, structured for Google Sheets connector compatibility.

Dataset	File	Shape	Looker Studio Use
GSC Queries	gsc_queries.csv	10 rows x 5 cols	Time-series: impressions, CTR, position
CWV Diagnostics	cwv_diagnostics.csv	5 rows x 5 cols	Scorecard: metric vs threshold
Schema Coverage	schema_coverage.csv	12 rows x 5 cols	Table: validation status by type
GEO Citation Trend	geo_citation_trend.csv	6 rows x 4 cols	Line chart: citation rate by platform
Technical Audit	seo_audit_checklist.csv	15 rows x 4 cols	Table: pass/fail by category

Recommended Looker Studio dashboard structure: Page 1 — GSC Performance (impressions, clicks, CTR, position trend); Page 2 — Technical Health (CWV scorecard, audit checklist); Page 3 — GEO/AEO Monitoring (citation rate by platform, trend). Full connection guide in P6 README.

Skills Demonstrated

Category	Specific Skills
Technical SEO	Crawlability audit, indexability, hreflang, canonical tags, sitemap management
Core Web Vitals	LCP/INP/CLS/FCP/TTFB measurement, PageSpeed Insights API, CrUX interpretation
Schema / Structured Data	JSON-LD authoring, Schema.org Validator, Google Rich Results Test, three-layer validation pipeline
GEO / AEO	Citation rate measurement, Princeton GEO benchmark methodology, multi-platform sampling (Google AI Overviews, Perplexity, ChatGPT, Bing)
Google Search Console	API integration, snapshot architecture, query/page performance analysis
Looker Studio	Export schema design, Google Sheets connector, BI dashboard structure
Python / Streamlit	Multi-page dashboard, st.download_button, st.line_chart, st.progress, data_disclosure component
Data transparency	Explicit real/snapshot/simulated labeling throughout; honest limitations documented

Limitations and Future Work

Known limitations

CWV values are simulated using PageSpeed Insights lab benchmarks; real CrUX field data requires sufficient real-user traffic, which Coffra (fictional brand) does not have. GEO citation monitoring is based on manual query sampling — inherently noisy, with approximately +/-14% margin of error at 48 queries per platform. No official Google AI Overviews citation API exists as of April 2026; monitoring requires manual spot-checking. Citation rate improvement may be partially attributable to factors beyond Schema (content freshness, concurrent backlink acquisition).

Future enhancements (v1.1+)

Automate GSC extraction via GitHub Actions + Google Cloud Run (weekly refresh). Replace CWV simulation with live CrUX API data as traffic grows. Automate Perplexity citation sampling via API. Increase query sample to 100 per platform (+/-10% margin of error). Design holdout experiment: Schema on test pages, withhold from matched control pages, measure citation rate difference — proper experimental GEO attribution. Add competitive benchmarking: Coffra citation rates vs 3 competitor coffee brands.

Connection to Other Projects

P6 closes the technical foundation loop across the entire Coffra portfolio:

Project	Relationship to P6
P1 — Marketing Automation	Email content drives organic traffic; GEO citations of email topics feed back into subject line testing (high-citation angles = higher open rates)
P2 — Marketing Dashboard	P6 is page 6 in the same Streamlit app; snapshot architecture identical to P2
P3 — Customer Segmentation	GSC query data can be segmented by persona (Connoisseur vs Daily Ritualist search behavior)
P4 — AEO Content Strategy	P4 built the Schema templates; P6 validates them, monitors their GEO impact, and measures the +41pp citation rate uplift
P5 — Attribution Modeling	As GEO-referred traffic grows, AI Overview / Perplexity becomes a measurable channel in P5's MMM — closing the attribution loop

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